

HAPPY WEEKEND TASK

MACHINE LEARNING DATA INVESTIGATION CHALLENGE

Number of Groups: 10

Difficulty: Challenging

Submission Deadline: Before Wednesday

Submission Method: GitHub Repository

1. GENERAL TASK INSTRUCTIONS

Your group is required to develop a complete Machine Learning solution for the assigned problem.

The purpose of this task is to test your understanding of:

- Data
- Data quality
- Data preparation
- Exploratory Data Analysis
- Feature engineering
- Machine Learning
- Model evaluation
- Model interpretation
- Data leakage
- Model deployment

You are expected to **investigate the data independently**.

Do not assume that the dataset is correctly prepared.

If you observe unusual model behavior, investigate the reason before accepting the result.

Requirements

1. Do not fabricate results.
2. Do not remove a variable simply because it improves or reduces accuracy without investigating why.
3. All conclusions must be supported by experiments.
4. Use appropriate validation and evaluation procedures.
5. The final solution must be reproducible.
6. Every group must submit the complete project to GitHub **before Wednesday**.

2. GROUP ALLOCATION

Group	Assignment
Groups 1–5	Synthetic Data Investigation
Group 6	Real Dataset
Group 7	Real Dataset
Group 8	Real Dataset
Group 9	Real Dataset
Group 10	Real Dataset

GROUP 1 — CUSTOMER RISK PREDICTION

Business Scenario

A financial company wants to identify customers who are likely to be high-risk.

Use the following code to generate your dataset.

Do not change the generation code before completing your initial investigation.

```
import numpy as np
import pandas as pd

np.random.seed(42)

n = 5000

income = np.random.lognormal(10, 2, n)
age = np.random.randint(18, 70, n)
transactions = np.random.randint(1, 100, n)

risk_score = (
    0.000001 * income
    + 0.02 * transactions
    - 0.01 * age
)

target = (risk_score > np.median(risk_score)).astype(int)

df = pd.DataFrame({
    "age": age,
    "income": income,
    "transactions": transactions,
    "risk_score": risk_score,
    "target": target
})

df.to_csv("group1_dataset.csv", index=False)
```

```
print(df.head())  
print(df.shape)  
print(df["target"].value_counts())
```

Questions / Requirements

1. Define the Machine Learning problem represented by the dataset.
2. Perform a complete data inspection before training any model.
3. Investigate the distribution and scale of every feature.
4. Train at least three different Machine Learning models.
5. Record training and validation/test performance separately.
6. Investigate any unusually high or low performance.
7. Determine whether every feature is appropriate for prediction.
8. Investigate relationships between the features and target.
9. Identify any variable that may provide information too directly related to the target.
10. Remove or transform any problematic feature only after explaining why.
11. Retrain the models and compare the results.
12. Perform cross-validation.
13. Perform error analysis.
14. Determine whether the final model can generalize to unseen data.
15. Save the final model.

GROUP 2 — MEDICAL RISK PREDICTION

Business Scenario

A healthcare organization wants to classify patients into two risk categories.

```
import numpy as np
import pandas as pd
np.random.seed(123)
n = 6000
age = np.random.randint(18, 85, n)
bmi = np.random.normal(27, 6, n)
blood_pressure = np.random.normal(125, 20, n)
glucose = np.random.normal(110, 35, n)
clinical_index = (
    0.04 * age
    + 0.08 * bmi
    + 0.03 * blood_pressure
    + 0.06 * glucose
)
risk = (clinical_index > np.percentile(clinical_index, 55)).astype(int)
df = pd.DataFrame({
    "age": age,
    "bmi": bmi,
    "blood_pressure": blood_pressure,
    "glucose": glucose,
    "clinical_index": clinical_index,
    "risk": risk
})
df.to_csv("group2_dataset.csv", index=False)
print(df.head())
print(df.info())
print(df["risk"].value_counts())
```

Questions / Requirements

1. Define the prediction problem and target variable.
2. Inspect the dataset for data-quality problems.

3. Perform exploratory data analysis.
4. Train at least three different classification models.
5. Record training, validation and test performance.
6. Investigate whether the model performance is unusually high.
7. Examine the relationship between every feature and the target.
8. Investigate whether any feature contains information derived from another feature or from the target-generation process.
9. Create a second experiment after removing any suspicious feature.
10. Compare the original and corrected experiments.
11. Investigate whether scaling changes model behavior.
12. Perform cross-validation and hyperparameter tuning.
13. Evaluate Accuracy, Precision, Recall, F1-score and ROC-AUC.
14. Perform error analysis.
15. Save and deploy the final model.

GROUP 3 — AGRICULTURAL DISEASE DETECTION

Business Scenario

An agricultural technology company wants to predict whether a crop observation belongs to a diseased or healthy class.

```
import numpy as np
import pandas as pd
np.random.seed(77)
n = 5000
temperature = np.random.normal(27, 5, n)
humidity = np.random.uniform(30, 100, n)
soil_moisture = np.random.uniform(5, 95, n)
rainfall = np.random.lognormal(3, 1.8, n)
environment_index = (
    0.03 * temperature
    + 0.02 * humidity
    + 0.04 * soil_moisture
    + 0.0001 * rainfall
)
disease = (environment_index > np.median(environment_index)).astype(int)
df = pd.DataFrame({
    "temperature": temperature,
    "humidity": humidity,
    "soil_moisture": soil_moisture,
    "rainfall": rainfall,
    "environment_index": environment_index,
    "disease": disease
})
df.to_csv("group3_dataset.csv", index=False)
print(df.describe())
print(df["disease"].value_counts())
```

Questions / Requirements

1. Define the Machine Learning task.
2. Inspect all variables and their distributions.
3. Investigate the relationship between environmental variables.
4. Train at least three classification algorithms.
5. Record training and test results.
6. Investigate whether the performance is realistic.
7. Investigate whether any feature is mathematically or logically derived from other predictors.
8. Test the effect of removing suspicious or redundant features.
9. Investigate feature correlation and multicollinearity.
10. Perform appropriate preprocessing.
11. Perform cross-validation.
12. Tune at least one model.
13. Perform error analysis.
14. Compare all experiments and identify the most reliable model.
15. Deploy the final system.

GROUP 4 — STUDENT DROPOUT PREDICTION

Business Scenario

A university wants to predict whether students are at risk of dropping out.

```
import numpy as np
import pandas as pd
np.random.seed(88)
n = 5000
attendance = np.random.uniform(30, 100, n)
study_hours = np.random.uniform(0, 15, n)
assignments = np.random.uniform(20, 100, n)
lms_activity = np.random.lognormal(5, 2, n)
academic_index = (
    0.04 * attendance
    + 0.10 * study_hours
    + 0.03 * assignments
    + 0.0005 * lms_activity
)
dropout = (academic_index < np.median(academic_index)).astype(int)
df = pd.DataFrame({
    "attendance": attendance,
    "study_hours": study_hours,
    "assignment_score": assignments,
    "lms_activity": lms_activity,
    "academic_index": academic_index,
    "dropout": dropout
})
df.to_csv("group4_dataset.csv", index=False)
print(df.head())
print(df.shape)
print(df["dropout"].value_counts())
```

Questions / Requirements

1. Define the problem and identify the target.
2. Perform data inspection and EDA.
3. Train at least three ML algorithms.
4. Record both training and unseen-data performance.
5. Investigate whether the performance is suspicious.
6. Investigate the construction of each feature.
7. Investigate whether one feature contains information from other features.
8. Repeat training after removing any suspicious variable.
9. Compare the results.
10. Investigate overfitting.
11. Investigate whether normalization/scaling changes the results.
12. Perform cross-validation and tuning.
13. Perform error analysis.
14. Explain which model should be used in practice and why.
15. Deploy the final model.

GROUP 5 — BUSINESS CUSTOMER CLASSIFICATION

Business Scenario

A company wants to identify customers belonging to a high-risk category.

```
import numpy as np
import pandas as pd
np.random.seed(99)
n = 5000
customer_age = np.random.randint(18, 80, n)
monthly_spend = np.random.lognormal(6, 2.5, n)
visits = np.random.randint(1, 80, n)
complaints = np.random.poisson(2, n)
customer_index = (
    0.02 * customer_age
    + 0.00001 * monthly_spend
    + 0.04 * visits
    - 0.08 * complaints
)
high_risk = (customer_index > np.median(customer_index)).astype(int)
df = pd.DataFrame({
    "customer_age": customer_age,
    "monthly_spend": monthly_spend,
    "visits": visits,
    "complaints": complaints,
    "customer_index": customer_index,
    "high_risk": high_risk
})
df.to_csv("group5_dataset.csv", index=False)
print(df.head())
print(df.describe())
print(df["high_risk"].value_counts())
```

Questions / Requirements

1. Define the Machine Learning problem.
2. Inspect the complete dataset.
3. Investigate numerical ranges and distributions.
4. Train at least three ML algorithms.
5. Compare training and test performance.
6. Investigate unusually high performance.
7. Investigate feature construction and feature relationships.
8. Determine whether any feature should be excluded and justify the decision.
9. Investigate the effect of normalization.
10. Investigate the effect of removing extreme observations.
11. Perform feature engineering.
12. Perform cross-validation and hyperparameter tuning.
13. Compare evaluation metrics.
14. Perform error analysis and model interpretation.
15. Save and deploy the final model.

GROUP 6 — REAL DATASET: Pima Indians Diabetes

Dataset

Kaggle:

<https://www.kaggle.com/datasets/vishvaachhatrara/missing-values-pima-indians-diabetes-data>

Questions / Requirements

1. Download the dataset and preserve the original raw file.
2. Inspect all columns, data types and dimensions.
3. Calculate missing-value percentages for every column.
4. Investigate whether zero values represent valid measurements or missing values.
5. Investigate duplicates and outliers.
6. Investigate class imbalance.
7. Determine an appropriate missing-value treatment strategy.
8. Compare at least two missing-value strategies.
9. Build a preprocessing pipeline.
10. Train at least four ML models.
11. Perform cross-validation and hyperparameter tuning.
12. Evaluate Accuracy, Precision, Recall, F1-score and ROC-AUC.
13. Perform error analysis.
14. Interpret the final model.
15. Deploy the final model.

GROUP 7 — REAL DATASET: PATIENT HEALTH RECORDS

Dataset

Patient Health Record Raw CSV

Kaggle:

<https://www.kaggle.com/datasets/mursaleen880/patient-health-record-raw-csv>

Questions / Requirements

1. Download and inspect every file/column provided.
2. Identify all missing values.
3. Identify invalid numerical values.
4. Identify inconsistent categorical values.
5. Identify duplicated or inconsistent records.
6. Investigate date and time fields.
7. Determine the prediction target.
8. Investigate possible target leakage.
9. Clean the dataset and justify every major cleaning decision.
10. Construct an ML-ready dataset.
11. Train at least four ML models.
12. Compare preprocessing approaches.
13. Perform cross-validation and hyperparameter tuning.
14. Perform complete model evaluation.
15. Perform error analysis.
16. Deploy the final model.

GROUP 8 — REAL DATASET: SALES AND MARKETING

Dataset

Sales and Marketing DataSet

Kaggle:

<https://www.kaggle.com/datasets/bhaskerpaul/sales-and-marketing-dataset>

Questions / Requirements

1. Download the raw dataset.
2. Identify every data-quality problem.
3. Calculate missing-value percentages.
4. Investigate outliers and extreme values.
5. Investigate inconsistent data types.
6. Investigate duplicated records.
7. Investigate class imbalance.
8. Determine which columns should be used as predictors.
9. Identify possible leakage variables.
10. Create at least five meaningful engineered features.
11. Perform feature selection.
12. Train at least four ML models.
13. Perform cross-validation and hyperparameter tuning.
14. Compare appropriate evaluation metrics.
15. Perform error analysis.
16. Interpret the final model.
17. Deploy the final model.

GROUP 9 — DATASET: RETAIL

Dataset

Retail Customer & Transaction Dataset

Kaggle:

<https://www.kaggle.com/datasets/raghavendragandhi/retail-customer-and-transaction-dataset>

Questions / Requirements

1. Download all relevant CSV files.
2. Inspect each dataset independently.
3. Identify primary keys and foreign keys.
4. Determine the relationships between tables.
5. Determine the correct merge/join strategy.
6. Record the number of rows before and after every merge.
7. Investigate duplicated records created by joining.
8. Investigate unmatched keys.
9. Investigate missing values introduced during merging.
10. Construct **ONE final dataset** for Machine Learning training.
11. Create an appropriate target variable.
12. Perform feature engineering using information from multiple source tables.
13. Train at least four ML models.
14. Perform feature selection.
15. Perform cross-validation and hyperparameter tuning.
16. Evaluate the final model.
17. Perform error analysis.
18. Document the complete data lineage from raw tables to final dataset.
19. Deploy the final model.

GROUP 10 — DATASET: BANKING / FRAUD

Dataset

Synthetic Global Bank Transactions Dataset

Kaggle:

<https://www.kaggle.com/datasets/mckenziemakwela/synthetic-global-bank-transactions-dataset>

Questions / Requirements

1. Download the customer and transaction datasets.
2. Inspect both datasets independently.
3. Identify the common key.
4. Verify key uniqueness and integrity.
5. Determine the relationship between customers and transactions.
6. Determine the correct join strategy.
7. Merge the datasets into **ONE training dataset**.
8. Verify row counts before and after merging.
9. Identify unmatched records.
10. Identify duplicate transactions or duplicated customer records.
11. Investigate missing values after the merge.
12. Investigate class imbalance.
13. Create transaction-level and customer-level features.
14. Train at least four classification models.
15. Perform feature selection.
16. Perform cross-validation and hyperparameter tuning.
17. Evaluate Precision, Recall, F1-score, ROC-AUC and an appropriate business metric.
18. Perform false-positive and false-negative analysis.
19. Save and deploy the final fraud-detection model.
20. Document the complete ETL and Machine Learning pipeline.

REQUIRED MACHINE LEARNING WORKFLOW — ALL GROUPS

Every group must complete the following:

1. Problem Definition
2. Data Collection / Generation
3. Data Inspection
4. Data Quality Assessment
5. Exploratory Data Analysis
6. Data Cleaning
7. Feature Engineering
8. Feature Selection
9. Train/Validation/Test Split
10. Data Preprocessing
11. Baseline Model
12. Model Training
13. Model Comparison
14. Hyperparameter Tuning
15. Cross-Validation
16. Final Evaluation
17. Error Analysis
18. Model Interpretation
19. Model Saving
20. Deployment

INVESTIGATION QUESTIONS — ALL GROUPS

Answer the following questions in your notebook:

1. What problem are you solving?
2. What type of Machine Learning problem is this?
3. What is the target variable?
4. What does each feature represent?
5. What problems exist in the raw data?
6. Which variables require preprocessing?
7. Which variables require transformation?
8. Are there outliers?
9. Are there duplicates?
10. Are there missing values?
11. Are there suspicious variables?
12. Could there be data leakage?
13. Is the target balanced?
14. What happens when the model is trained using the raw data?
15. What happens after preprocessing?
16. Is the training result realistic?
17. Does the test result support the training result?
18. Which model performs best?
19. Why does that model perform best?
20. Which errors are most important?
21. Can the model generalize to unseen data?
22. What evidence supports your conclusion?
23. What are the limitations of your final solution?

IMPORTANT

You must investigate your data and provide evidence.

GITHUB SUBMISSION — BEFORE WEDNESDAY

Every group **MUST** submit a GitHub repository before Wednesday.

The repository must contain:

- Jupyter Notebook
- Raw data or data-generation code
- Processed dataset or reproducible preprocessing code
- Saved final model
- Deployment application
- requirements.txt
- README.md
- Screenshots of the application
- Final model results table
- GitHub repository link

REQUIRED GITHUB STRUCTURE

groupXX-ml-weekend-task/

```
|
|
├── data/
|   ├── raw/
|   └── processed/
|
|
├── notebooks/
|   └── groupXX_project.ipynb
|
|
├── src/
|   └── data_generation.py
|
|
├── models/
|   └── best_model.joblib
|
|
├── app/
|   └── app.py
|
|
├── requirements.txt
├── README.md
└── screenshots/
```

For Groups 1–5, the synthetic-data generation code must be included.

FINAL SUBMISSION QUESTIONS

1. What did you initially observe when training your first model?
2. Was the initial result trustworthy?
3. What investigation did you perform to understand the result?
4. What was the actual cause of the unusual model behavior, if any?
5. How did you prove your explanation?
6. What changed after correcting the problem?
7. Which preprocessing decisions were necessary?
8. Which model was finally selected?
9. Why was it selected?
10. How did you test generalization?
11. How did you investigate data leakage?
12. What are the main limitations of your system?
13. How would you improve the system for production use?

FINAL CHALLENGE

Your grade will depend not only on the final accuracy, but on your ability to investigate the data, identify hidden problems, explain unexpected model behavior, justify your preprocessing and modeling decisions, and demonstrate that your final model can generalize to unseen data.

NB:

Submit your complete GitHub repository before Wednesday. And Invite Accessor to your github repository.